Rafael Angelo Christianto

HACK THE BOX SHERLOCK

ROMCOM

Here we are presented with an .evtx file which is for (Windows Event Log) files upon unzipping the unit42.zip.

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Unit42]
└─$ ls
Microsoft-Windows-Sysmon-Operational.evtx  unit42.zip
```

To read the Windows Event Logs better, I transformed it to a json file using the Chainsaw command utility

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Unit42]
└─$ chainsaw dump Microsoft-Windows-Sysmon-Operational.evtx --json > output.json

By WithSecure Countercept (@FranticTyping, @AlexKornitzer)

[+] Dumping the contents of forensic artefacts from: Microsoft-Windows-Sysmon-Operational.evtx (extensions: *)
[+] Loaded 1 forensic artefacts (1.1 MiB)
[+] Done
```

Then it produced a parsed version called output.json.

Then from here are the questions from HackTheBox.

1. How many Event logs are there with Event ID 11?

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Unit42]
└─$ strings output.json | grep -c '"EventID": 11'
56
```

   I then used the strings for the output.json file and grep for string Event Id of 11, and counted it by using the -c flag and we get the answer of **56**.

2. Whenever a process is created in memory, an event with Event ID 1 is recorded with details such as command line, hashes, process path, parent process path, etc. This information is very useful for an analyst because it allows us to see all programs executed on a system, which means we can spot any malicious processes being executed. What is the malicious process that infected the victim's system?

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Unit42]
└─$ strings output.json | grep -iE '"EventID": 1'
    "EventID": 11,
    "EventID": 11,
    "EventID": 11,
    "EventID": 11,
    "EventID": 15,
```

I tried to use grep, but it cannot show the full attributes, therefore I moved to jq to search under specific attribute groups until EventID == 1.

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Unit42]
└─$ jq '.[] | select(.Event.System.EventID == 1)' output.json
{
  "Event_attributes": {
    "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
  },
  "Event": {
    "System": {
      "Provider_attributes": {
        "Name": "Microsoft-Windows-Sysmon",
        "Guid": "5770385F-C22A-43E0-BF4C-06F5698FFBD9"
      },
      "EventID": 1,
      "Version": 5,
      "Level": 4,
      "Task": 1,
      "Opcode": 0,
      "Keywords": "0x8000000000000000",
      "TimeCreated_attributes": {
        "SystemTime": "2024-02-14T03:41:45.305882Z"
```

I then searched for an oddly named file

```
  "Description": "Photo and vn Installer",
  "Product": "Photo and vn",
  "Company": "Photo and Fax Vn",
  "OriginalFileName": "Fattura 2 2024.exe",
  "CommandLine": "\"C:\\Users\\CyberJunkie\\Downloads\\Preventivo24.02.14.exe.exe\\" ",
  "CurrentDirectory": "C:\\Users\\CyberJunkie\\Downloads\\",
  "User": "DESKTOP-887GK2L\\CyberJunkie",
```

Then I found this, highlighted in the screenshot above, and I reduced the "\" into 1 for each, instead of 2. And the answer is **C:\Users\CyberJunkie\Downloads\Preventivo24.02.14.exe.exe**.

3. Which Cloud drive was used to distribute the malware?

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Unit42]
└─$ jq '.[] | select(.Event.System.EventID == 22)' output.json
{
  "Event_attributes": {
    "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
  },
  "Event": {
    "System": {
      "Provider_attributes": {
        "Name": "Microsoft-Windows-Sysmon",
        "Guid": "5770385F-C22A-43E0-BF4C-06F5698FFBD9"
      },
      "EventID": 22,
      "Version": 5,
      "Level": 4,
      "Task": 22,
      "Opcode": 0,
      "Keywords": "0x8000000000000000",
      "TimeCreated_attributes": {
        "SystemTime": "2024-02-14T03:41:26.444119Z"
      },
      "EventRecordID": 118747,
      "Correlation": null,
      "Execution_attributes": {
        "ProcessID": 3028,
        "ThreadID": 4452
      },
      "Channel": "Microsoft-Windows-Sysmon/Operational",
      "Computer": "DESKTOP-887GK2L",
      "Security_attributes": {
```

Here for Cloud related activities usually it has the Event ID of 22, so I will filter on that again.

And after counting it there are only 3 events with Event ID 22.

```
File  Actions  Edit  View  Help
      "TimeCreated_attributes": {
        "SystemTime": "2024-02-14T03:41:58.764837Z"
      },
      "EventRecordID": 118906,
      "Correlation": null,
      "Execution_attributes": {
        "ProcessID": 3028,
        "ThreadID": 4452
      },
      "Channel": "Microsoft-Windows-Sysmon/Operational",
      "Computer": "DESKTOP-887GK2L",
      "Security_attributes": {
        "UserID": "S-1-5-18"
      }
    },
    "EventData": {
      "RuleName": "-",
      "UtcTime": "2024-02-14 03:41:56.955",
      "ProcessGuid": "817BDDF3-3684-65CC-2D02-000000001900",
      "ProcessId": 10672,
      "QueryName": "www.example.com",
      "QueryStatus": "0",
      "QueryResults": "::ffff:93.184.216.34;199.43.135.53;2001:500:8f::53;199.43.133.53;2001:500:8d::53;",
      "Image": "C:\\Users\\CyberJunkie\\Downloads\\Preventivo24.02.14.exe.exe",
      "User": "DESKTOP-887GK2L\\CyberJunkie"
    }
  }
}
```

In the screenshot above, we see the answer to the previous question, which is the Preventio executable. Then near that we see a dropbox link in the other 2 events above it, where dropbox is a sync document sharing platform similar to Google Drive, so that has to be the answer to the 3rd question, which is **dropbox**.

4. For many of the files it wrote to disk, the initial malicious file used a defense evasion technique called Time Stomping, where the file creation date is changed to make it appear older and blend in with other files. What was the timestamp changed to for the PDF file?


```
    }
  },
  "EventData": {
    "RuleName": "technique_id=T1070.006,technique_name=Timestomp",
    "UtcTime": "2024-02-14 03:41:58.404",
    "ProcessGuid": "817BDDF3-3684-65CC-2D02-000000001900",
    "ProcessId": 10672,
    "Image": "C:\\Users\\CyberJunkie\\Downloads\\Preventivo24.02.14.exe.exe",
    "TargetFilename": "C:\\Users\\CyberJunkie\\AppData\\Roaming\\Photo and Fax Vn\\Photo and vn 1.1.2\\install\\F97891C\\TempFolder\\~.pdf",
    "CreationUtcTime": "2024-01-14 08:10:06.029",
    "PreviousCreationUtcTime": "2024-02-14 03:41:58.404",
    "User": "DESKTOP-887GK2L\\CyberJunkie"
  }
}
```


Here we see a pdf time called "~.pdf", we can see 2 UTC times in the screenshot above, the 2024-02-14 03:41:58 is the time when the file was actually created, but then the hacker changed it to **2024-01-14 08:10:**06 to make it look more legit

5. The malicious file dropped a few files on disk. Where was "once.cmd" created on disk? Please answer with the full path along with the filename.


```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Unit42]
└─$ strings output.json | grep -iE "once.cmd"
      "TargetFilename": "C:\\Users\\CyberJunkie\\AppData\\Roaming\\Photo and Fax Vn\\Photo and vn 1.1.2\\install\\F97891C\\WindowsVolume\\Games\\
once.cmd",
      "TargetFilename": "C:\\Users\\CyberJunkie\\AppData\\Roaming\\Photo and Fax Vn\\Photo and vn 1.1.2\\install\\F97891C\\WindowsVolume\\Games\\
once.cmd",
      "TargetFilename": "C:\\Games\\once.cmd",
      "TargetFilename": "C:\\Users\\CyberJunkie\\AppData\\Roaming\\Photo and Fax Vn\\Photo and vn 1.1.2\\install\\F97891C\\WindowsVolume\\Games\\
once.cmd",
```


I searched output.json for the keyword "once.cmd" using the grep CLI utility, and we found several, but we found the full path, and I just have to reduce from "\\" to "\". So the final answer is:

C:\Users\CyberJunkie\AppData\Roaming\Photo and Fax Vn\Photo and vn 1.1.2\install\F97891C\WindowsVolume\Games\once.cmd

6. The malicious file attempted to reach a dummy domain, most likely to check the internet connection status. What domain name did it try to connect to?

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Unit42]
└─$ jq '.[] | select(.Event.System.EventID == 22)' output.json
{
  "Event_attributes": {
    "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
  },
  "Event": {
    "System": {
      "Provider_attributes": {
        "Name": "Microsoft-Windows-Sysmon",
        "Guid": "5770385F-C22A-43E0-BF4C-06F5698FFBD9"
      },
      "EventID": 22,
      "Version": 5,
      "Level": 4,
      "Task": 22,
      "Opcode": 0,
      "Keywords": "0x8000000000000000",
      "TimeCreated_attributes": {
        "SystemTime": "2024-02-14T03:41:26.444119Z"
      },
      "EventRecordID": 118747,
      "Correlation": null,
      "Execution_attributes": {
        "ProcessID": 3028,
        "ThreadID": 4452
      },
      "Channel": "Microsoft-Windows-Sysmon/Operational",
      "Computer": "DESKTOP-887GK2L",
      "Security_attributes": {
```

I again used jq to search for EventID 22 through output.json and near the bottom parts, I found an oddly sounding domain that anyone can enter to test internet connection.

```
"EventData": {
  "RuleName": "-",
  "UtcTime": "2024-02-14 03:41:56.955",
  "ProcessGuid": "817BDDF3-3684-65CC-2D02-000000001900",
  "ProcessId": 10672,
  "QueryName": "www.example.com",
  "QueryStatus": "0",
  "QueryResults": "::ffff:93.184.216.34;199.43.135.53;2001:500:8f::53;199.43.133.53;2001:500:8d::53;",
  "Image": "C:\\Users\\CyberJunkie\\Downloads\\Preventivo24.02.14.exe.exe",
  "User": "DESKTOP-887GK2L\\CyberJunkie"
}
```

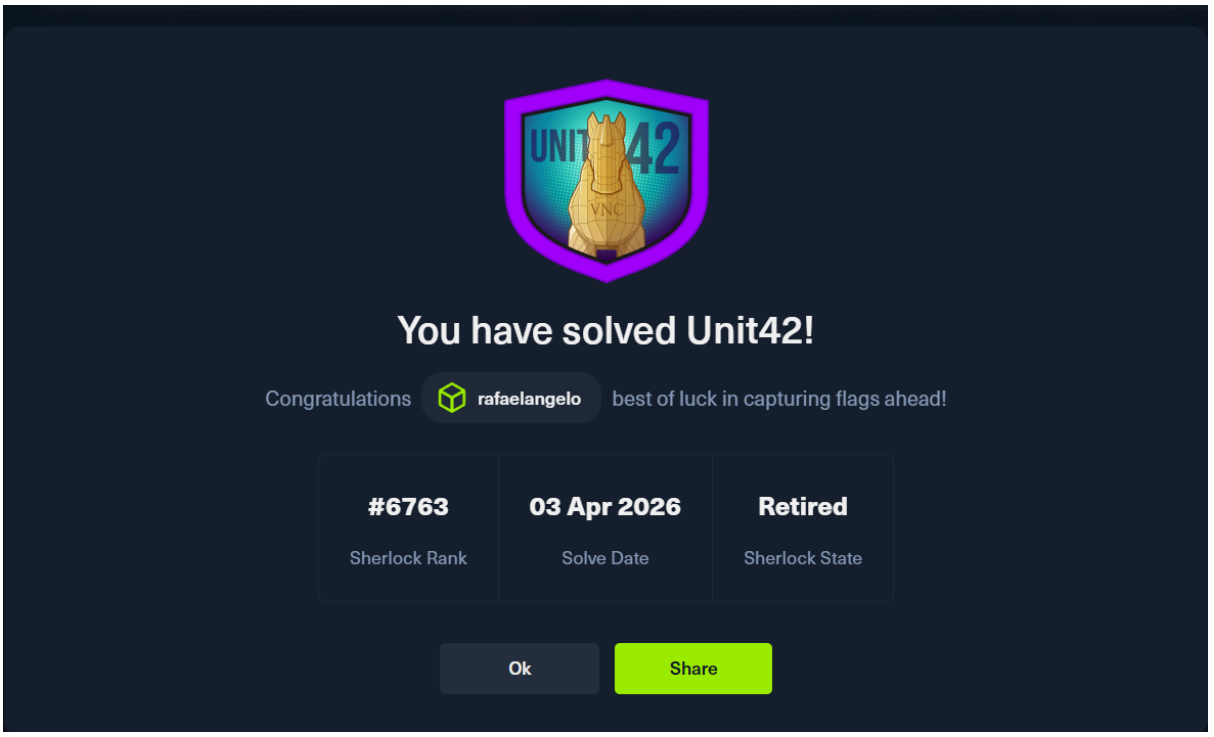
And from there, I know the answer is **www.example.com.**

7. Which IP address did the malicious process try to reach out to?

```
    },
    "EventData": {
      "RuleName": "technique_id=T1036,technique_name=Masquerading",
      "UtcTime": "2024-02-14 03:41:57.159",
      "ProcessGuid": "817BDDF3-3684-65CC-2D02-000000001900",
      "ProcessId": 10672,
      "Image": "C:\\Users\\CyberJunkie\\Downloads\\Preventivo24.02.14.exe.exe",
      "User": "DESKTOP-887GK2L\\CyberJunkie",
      "Protocol": "tcp",
      "Initiated": true,
      "SourceIsIpv6": false,
      "SourceIp": "172.17.79.132",
      "SourceHostname": "-",
      "SourcePort": 61177,
      "SourcePortName": "-",
      "DestinationIsIpv6": false,
      "DestinationIp": "93.184.216.34",
      "DestinationHostname": "-",
      "DestinationPort": 80,
      "DestinationPortName": "-"
    }
  }
}
```


After looking at the same event that delivered the Preventio executable file which was discussed in an earlier question, we see a destination IP, so that has to be the IP that the malicious process tried to reach out to which is **93.184.216.34**.

8. The malicious process terminated itself after infecting the PC with a backdoored variant of UltraVNC. When did the process terminate itself?


```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Unit42]
└─$ jq '.[] | select(.Event.System.EventID == 5)' output.json
{
  "Event_attributes": {
    "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
  },
  "Event": {
    "System": {
      "Provider_attributes": {
        "Name": "Microsoft-Windows-Sysmon",
        "Guid": "5770385F-C22A-43E0-BF4C-06F5698FFBD9"
      },
      "EventID": 5,
      "Version": 3,
      "Level": 4,
      "Task": 5,
      "Opcode": 0,
      "Keywords": "0x8000000000000000",
      "TimeCreated_attributes": {
        "SystemTime": "2024-02-14T03:41:58.799651Z"
      },
      "EventRecordID": 118907,
      "Correlation": null,
      "Execution_attributes": {
        "ProcessID": 3028,
        "ThreadID": 4412
      },
      "Channel": "Microsoft-Windows-Sysmon/Operational",
      "Computer": "DESKTOP-887GK2L",
      "Security_attributes": {
        "UserID": "S-1-5-18"
      }
    },
    "EventData": {
      "RuleName": "-",
      "UtcTime": "2024-02-14 03:41:58.795",
      "ProcessGuid": "817BDDF3-3684-65CC-2D02-000000001900",
      "ProcessId": 10672,
      "Image": "C:\\Users\\CyberJunkie\\Downloads\\Preventivo24.02.14.exe.exe",
      "User": "DESKTOP-887GK2L\\CyberJunkie"
    }
  }
```

For this question, I used the jq tool again but this time with Event ID 5 filter, and we only see 1 event, here we see the time the process self terminates, therefore it is also the answer, which is **2024-02-14 03:41:58**.